

```
port      = ssh
filter    = sshd
logpath   = /var/log/auth.log
maxretry  = 5
bantime   = 600
```

Save the file and restart Fail2ban for the changes to take effect:

```
sudo systemctl restart fail2ban.service
```

Enable Fail2ban at system startup:

```
sudo systemctl enable fail2ban.service
```

Install and enable mod_security:

Mod_security is a Web application firewall (WAF) that can be installed as an optional module for Apache. It can be used to protect the Web server from numerous attacks, such as SQL injection, session hijacking, cross-site scripting, bad user agents and many others. To install and enable mod_security, run the following commands:

```
sudo apt-get install libapache2-mod-security
sudo a2enmod security2
```

OWASP ModSecurity is another utility to protect Apache from DoS and DDoS attacks. The module also works with IP addresses, identifying those from which suspicious requests occur and placing them in the blacklist.

After installation, you must configure the module and enable the OWASP ModSecurity Core Rule Set (CRS):

```
sudo mv /etc/modsecurity/modsecurity.conf-recommended /etc/modsecurity/modsecurity.conf
```

Then open `/etc/modsecurity/modsecurity.conf` and add the following parameters:

```
SecRuleEngine On
SecResponseBodyAccess Off
SecRequestBodyLimit 8388608
SecRequestBodyNoFilesLimit 131072
```

```
SecRequestBodyInMemoryLimit 262144
```

Save and close the file. Delete the current CRS and load the OWASP CRS using the following commands:

```
sudo mv /etc/modsecurity/modsecurity.conf-recommended /etc/modsecurity/modsecurity.conf
Install and Enable mod_evasive
```

Mod_evasive is an Apache module that can be used to protect a Web server from DoS (Denial of Service), DDoS (Distributed Denial of Service) and brute force attacks. To install mod_evasive on the server, run the following command:

```
sudo apt-get install libapache2-mod-evasive
```

Open the default configuration file in `/etc/apache2/mods-enabled/evasive.conf` and change the settings to look like the following:

```
<IfModule mod_evasive20.c>
<IfModule mod_evasive20.c>
DOSPageCount 5
DOSSiteCount 50
DOSPageInterval 1
DOSSiteInterval 1
DOSBlockingPeriod 600
DOSLogDir "/var/log/mod_evasive"
</IfModule>
```

Save and close the file. Create a directory for the log files:

```
sudo mkdir /var/log/mod_evasive
sudo chown -R www-data: /var/log/mod_evasive
```

Restart Apache:

```
sudo systemctl restart apache2.service
```

Data protection

By default, Apache stores a large amount of information that requires protection; so to ensure the security of the Web server,

confidential data is best hidden. This can be done via the `conf` file. First, create a `custom.conf` file in the `conf-available` folder of the Apache Web server and paste the following lines there:

```
***
ServerTokens Prod
ServerSignature Off
TraceEnable Off
Options all -Indexes
Header unset ETag
Header always unset X-Powered-By
FileETag None
***
```

After that, enable the configuration with the `sudo a2enconf custom.conf` command and restart Apache.

MySQL protection

The first thing to do to protect the MySQL service is to run the `mysql_secure_installation` script.

The script will help you complete important security tasks like setting up the root password, disabling remote administrator login, and deleting anonymous users.

Disabling MySQL remote access:

If you do not perform remote operations on your MySQL server, disable remote access to the service. You can do this by editing the `/etc/mysql/mysql.conf.d/mysqld.cnf` file and changing the bind address to `127.0.0.1`.

Restart the service for the changes to take effect using the following script:

```
sudo systemctl restart mysql.service
```

Creating individual MySQL

users: Another thing you should consider is creating separate MySQL users for each database and application.

Log in to MySQL as root:

```
mysql -u root -p
```

You can create a MySQL database and grant all privileges to a new user using the following commands: